

INSTCHARVOICE: GROUNDING NATURAL-LANGUAGE INSTRUCTIONS FOR CHARACTER-LEVEL CONTROL IN TEXT-TO-SPEECH

Sihang Nie^{1*}, Xueru Li^{1*}, Xiaofen Xing^{1†}, Deyi Tuo², Cheng-Bin Jin², Jingyuan Xing¹, Jinxin Ji³

¹South China University of Technology, ²Huya Inc., ³The Hongkong Polytechnic University

ABSTRACT

Instruction-based text-to-speech (ITTS) systems enable natural-language control of expressive speech generation, but often offer limited transparency and fine-grained control over individual text units. Character-level controllable TTS systems provide explicit acoustic control, yet typically rely on user-specified acoustic attributes. To bridge this gap, we propose InstCharVoice, a unified framework that grounds natural-language instructions in character-level acoustic control. We first construct grounded instruction annotations on the WordVoice-5A-zh corpus using Qwen3-Omni. With this supervision, we train an autoregressive model to identify instruction-relevant characters and predict their acoustic attributes before generating the corresponding speech tokens. Keyword prediction and grounding-aware loss weighting help the model focus on instruction-relevant characters and attributes. Experiments show improved instruction following and keyword-level acoustic control over representative ITTS systems, with competitive speech naturalness and explicit character-level controllability. Audio samples are available at <https://xxh333.github.io/instcharvoice-demo/>.

Index Terms— Instruction-based TTS, instruction grounding, character-level control

1. INTRODUCTION

Neural text-to-speech (TTS) systems have achieved impressive expressiveness in zero-shot voice cloning. Benefiting from large-scale speech data and generative modeling, these systems can synthesize speech with high naturalness and speaker similarity [1, 2, 3, 4]. This progress has increasingly shifted research attention toward expressive instruction-based speech generation [5, 6, 7, 8, 9].

Despite this progress, existing instruction-based TTS (ITTS) systems often provide limited transparency and fine-grained control. Many systems directly generate speech from text and instructions without explicitly identifying which text units should be modified or which acoustic attributes should realize the requested changes [7, 10]. Consequently, the correspondence among instruction phrases, target text units, and acoustic variations remains implicit. Moreover, existing ITTS systems typically emphasize utterance-level or segment-level characteristics, leaving precise control over individual characters less explored [11, 12].

Fine-grained controllable TTS offers a complementary solution by representing local acoustic variations with explicit character-level attributes [12, 13]. Such representations enable precise and inspectable control over individual text units. However, these methods typically rely on user-specified acoustic attributes, which can

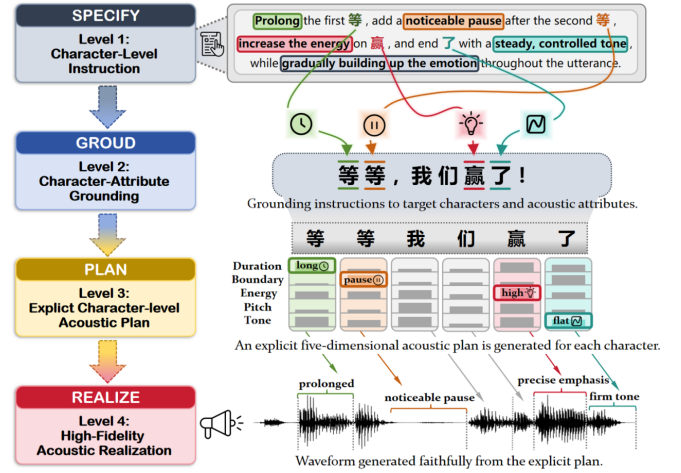


Fig. 1. Motivation and overview of InstCharVoice. The framework bridges natural-language instructions and explicit character-level acoustic control through instruction grounding and acoustic planning.

become cumbersome to configure as the utterance length and the number of controllable attributes increase. This attribute-based interface is less intuitive than expressing the desired changes in natural language. An important gap therefore remains between natural-language instruction control and explicit character-level acoustic realization.

To bridge this gap, we propose **InstCharVoice**, an ITTS framework for fine-grained character-level control. As illustrated in Fig. 1, InstCharVoice connects instruction specification, character-level grounding, acoustic planning, and speech realization. It identifies instruction-relevant characters and predicts five acoustic attributes—duration, boundary, energy, pitch, and tone—for each character before generating its corresponding speech tokens. To provide supervision for this grounded modeling process, we develop an LLM-assisted annotation pipeline that leverages an open-source multimodal language model to generate natural-language instructions and character-level grounding labels from text, speech, and acoustic attributes.

Our contributions are summarized as follows. 1) We develop an LLM-assisted annotation pipeline that links natural-language instructions to target characters and acoustic attributes, providing explicit supervision for instruction grounding. 2) We propose **InstCharVoice**, a unified framework that bridges natural-language instructions and speech generation through character-level acoustic planning, with keyword prediction and grounding-aware loss weighting to strengthen localized control. 3) Experiments show

*Work conducted when the author was intern at Huya Inc.

†Corresponding author.

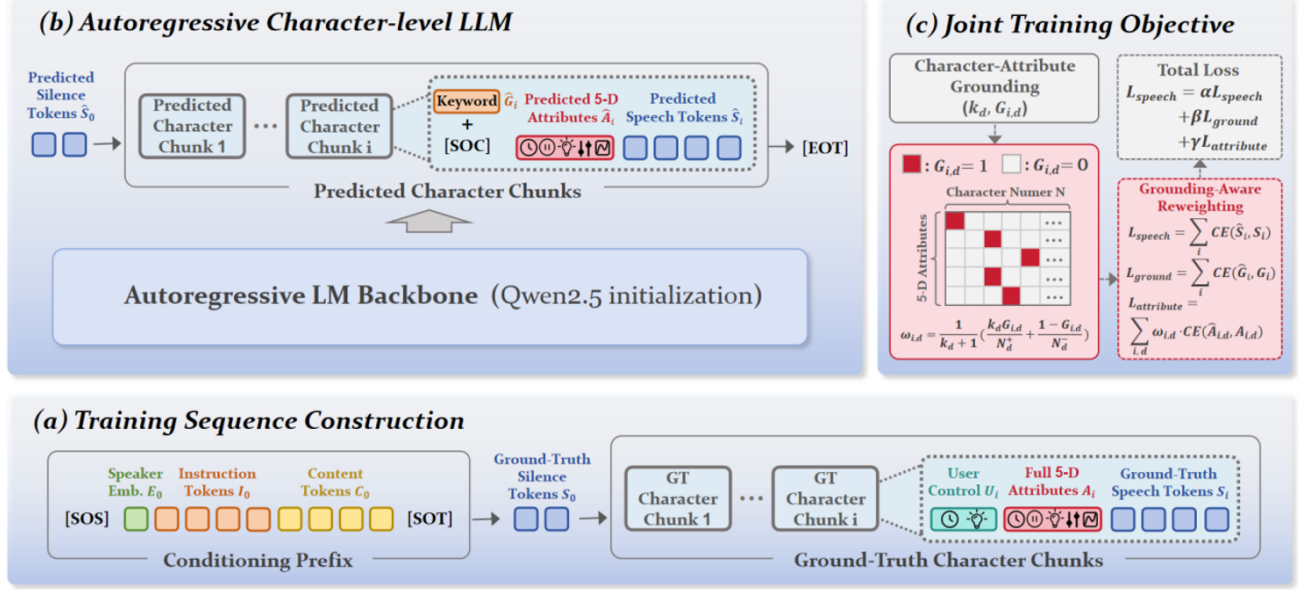


Fig. 2. Model architecture of InstCharVoice, including (a) training sequence construction, (b) autoregressive modeling of character-level speech chunks, and (c) grounding-aware training objective.

Table 1. Statistics of the character-level grounded instruction annotations. Dur, Bnd, Eng, Pit, and Ton denote duration, boundary, energy, pitch, and tone, respectively.

Statistic	Utterances	Keywords	Dur	Bnd	Eng	Pit	Ton
Count	2143k	5978k	609k	3882k	2830k	202k	901k

improved instruction following and keyword-level acoustic control over representative ITTS systems, while maintaining competitive speech naturalness and character-level controllability.

2. METHODOLOGY

2.1. Character-Level Grounded Instruction Annotation

We construct character-level grounded instruction annotations on WordVoice-5A-zh [12], a corpus derived from LEMAS [14]. The dataset contains approximately 2,546 hours of Chinese speech, with transcriptions and character-level annotations for duration, energy, pitch, boundary, and tone. We develop a three-stage annotation pipeline based on Qwen3-Omni-30B-A3B [15]¹.

First, Qwen3-Omni generates natural-language instructions from each speech-text pair to describe utterance-level expressive and acoustic characteristics. We then provide the character-level acoustic attributes as explicit cues to refine the descriptions and correct inconsistencies. Finally, salient local acoustic patterns, such as prolonged duration or prominent energy, are extracted from the character-level annotations. Guided by these patterns, the model further revises and supplements the instructions with localized acoustic requirements and assigns grounding labels to the corresponding character-attribute pairs. Only pairs addressed by the final instructions are labeled as grounded. The resulting annotations support

both utterance-level instruction following and character-level acoustic grounding. Table 1 summarizes the annotation statistics.

2.2. Character-Level Instruction-to-Speech Modeling

As illustrated in Fig. 2, InstCharVoice models instruction-based speech synthesis as autoregressive generation over character-level speech chunks. For each character, the model predicts its grounding label and acoustic attributes before generating the aligned speech tokens. This formulation supports keyword-aware instruction grounding, interpretable acoustic planning, and optional character-level user control.

Let I_0 , C_0 , and E_0 denote the instruction-token sequence, content-token sequence, and speaker embedding, respectively. The speaker embedding is extracted from the reference speech using a pre-trained voiceprint model [16]². The conditioning prefix is constructed as $\mathbf{P} = [\text{[SOS]}, E_0, I_0, C_0, \text{[SOT]}]$. It is followed by the leading silence tokens $\mathbf{S}_0$ and the ground-truth character chunks. For the i -th character, the chunk is represented as $\mathbf{X}_i^{\text{gt}} = [U_i, A_i, S_i]$, where U_i contains the user-specified acoustic attributes, A_i is the combined embedding of all five acoustic attributes, and S_i is the speech-token sequence segmented according to the character duration. During training, attributes in U_i are randomly provided or masked, allowing the model to predict the complete five-dimensional attributes from partial user specifications. Thus, the training sequence is $\mathbf{X}^{\text{gt}} = [\mathbf{P}, \mathbf{S}_0, \mathbf{X}_1^{\text{gt}}, \dots, \mathbf{X}_N^{\text{gt}}, \text{[EOT]}]$.

During inference, the model autoregressively predicts the leading silence tokens $\hat{\mathbf{S}}_0$ and character-level speech chunks. The speech-token head predicts $[\text{SOC}]$ to mark the transition to a new character chunk, while an auxiliary grounding head predicts $\hat{G}_i$ from the same hidden state, indicating whether the character belongs to an instruction-grounded keyword. Grounding prediction provides auxiliary supervision during training and is not fed back

¹<https://www.modelscope.cn/models/Qwen/Qwen3-Omni-30B-A3B-Instruct>

²https://www.modelscope.cn/models/iic/speech_campplus_sv_zh-cn_3dspeaker_16k

into the generation sequence. For each character, five attribute-specific heads predict the complete acoustic attributes $\hat{A}_i$ in parallel from a shared hidden state, conditioned on any attributes provided in U_i . Unspecified attributes are inferred from the instruction and generation context. The model then autoregressively generates the aligned speech tokens $\hat{S}_i$ until the next [SOC] or the sequence-ending token [EOT]. The generated attributes and speech tokens are fed into WordVoice-FM [12] to synthesize the output waveform.

2.3. Training Objectives

As illustrated in Fig. 2(c), InstCharVoice-LLM is jointly trained with three cross-entropy objectives: speech-token prediction, character-level grounding prediction, and five-dimensional acoustic attribute prediction. We adopt uncertainty-based loss weighting [17] to balance these objectives using learnable task uncertainties.

Let $\mathcal{D}$ denote the five acoustic attributes: duration, boundary, energy, pitch, and tone. For classification-based prediction, duration is represented by the number of aligned speech tokens, while energy and pitch are each uniformly quantized into 20 bins over their respective ranges. Boundary and tone use five and seven categories, respectively, following WordVoice-5A [12]. For the i -th character and attribute $d \in \mathcal{D}$, the grounding-aware attribute loss is defined as

$$\mathcal{L}_{\text{attribute}} = \sum_{i=1}^N \sum_{d \in \mathcal{D}} \omega_{i,d} \text{CE}(\hat{A}_{i,d}, A_{i,d}), \quad (1)$$

$$\omega_{i,d} = \frac{1}{k_d + 1} \left(\frac{k_d G_{i,d}}{N_d^+} + \frac{1 - G_{i,d}}{N_d^-} \right). \quad (2)$$

Here, $A_{i,d}$ and $\hat{A}_{i,d}$ denote the ground-truth label and predicted distribution, respectively; $G_{i,d} \in \{0, 1\}$ indicates whether the character-attribute pair is grounded by the instruction; and N_d^+ and N_d^- count the grounded and non-grounded pairs for attribute d within the utterance. Attributes with no grounded pairs are trained without grounding-aware reweighting. The factor k_d controls the relative contribution of grounded and non-grounded pairs after group-wise normalization. We set $k_d = 5$ for boundary and energy, and $k_d = 3$ for the remaining attributes.

We initialize the model from Qwen2.5-0.5B [18] and use the CosyVoice3 [19] tokenizer to extract speech tokens. The model is trained with Adam at a fixed learning rate of 2×10^{-5} on 8 NVIDIA A800 GPUs. Training consists of two stages: the model is first trained for five epochs without grounding-aware reweighting, focusing on global instruction-to-speech modeling and character-level acoustic attribute modeling for speech generation; the reweighting scheme is then enabled, and training continues for three additional epochs to further emphasize instruction-relevant acoustic attributes.

3. EXPERIMENTS

3.1. Evaluation of Grounded Instruction Annotations

We conduct a blind human evaluation to assess instruction-speech consistency and the benefit of character-level acoustic cues. Ten native Chinese speakers each evaluate 40 randomly sampled utterances from WordVoice-5A-zh [12]. For each utterance, the same listener evaluates two instruction annotations: *w/o cues*, generated from the speech-text pair in the first annotation stage, and *with cues*, refined using character-level acoustic measurements in the second stage. Listeners judge whether the acoustic characteristics described in each instruction are supported by the speech, using three categories: *inconsistent*, *uncertain*, and *consistent*.

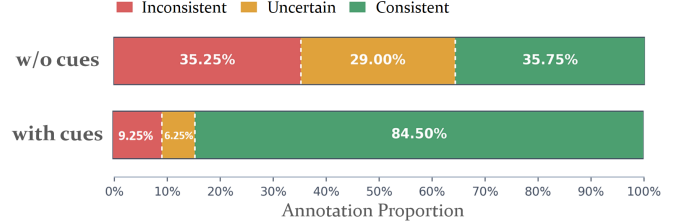


Fig. 3. Blind evaluation of instruction annotation consistency.

As shown in Fig. 3, adding character-level acoustic cues increases consistent judgments from 35.75% to 84.50%, while reducing inconsistent judgments from 35.25% to 9.25% and uncertain judgments from 29.00% to 6.25%. These results indicate that explicit acoustic measurements help the multimodal model produce descriptions that better match the speech, providing a more reliable instruction basis for subsequent local grounding.

3.2. Main Comparison

We evaluate InstCharVoice against three representative ITTS systems: **FireRedTTS3** [9], **Qwen3-TTS** [20], and **CosyVoice3** [19]. We follow the data split of WordVoice-5A and use the same evaluation texts and reference speech across the main comparison, ablation study, and explicit character-level control comparison.

For subjective evaluation, we randomly select 14 instruction-text pairs from the test set and one reference speech sample for each pair. All systems use the same instruction-text pairs and reference speech. Since the selected Qwen3-TTS and FireRedTTS3 models do not support voice cloning in instruction mode, we convert their outputs to the target speaker using Seed-VC [21]. A total of 27 native listeners rate the synthesized speech on a 5-point scale for naturalness (NMOS), instruction following (IMOS), and speaker similarity (SMOS). We report MOS [22] with 95% confidence intervals.

For objective evaluation, we use the first 30% of each ground-truth utterance as reference speech and compare the synthesized speech with the corresponding ground truth. We also include the ground-truth condition and an InstCharVoice variant without instruction input, denoted as *w/o Instruction*. Speech quality and intelligibility are measured using DNSMOS [23] and character error rate (CER) measured by Qwen3-ASR [24]. To assess localized acoustic control, we extract character-level attributes using the WordVoice-5A annotation pipeline [12] and compare them with the ground-truth annotations. Each metric is computed only on character-attribute pairs grounded by the instruction. We use the prefix “k” to denote keyword-level evaluation, followed by D, E, P, B, or T for duration, energy, pitch, boundary, or tone. Mean absolute error (MAE) is computed for duration, energy, and pitch using the normalized values from the original pipeline. Boundary comprises five pause-duration categories, while tone comprises seven categories derived from quadratic fits to within-character pitch contours. For both boundary and tone, we use a relaxed error rate (RER), which treats a prediction within one category of the target as correct. The nonzero acoustic errors in the GroundTruth condition arise from slight differences in timestamp extraction between the evaluation pipeline and the original annotation pipeline.

As shown in Tables 2 and 3, InstCharVoice achieves the highest mean NMOS and IMOS, the highest DNSMOS, and the lowest errors across all five keyword-level acoustic metrics among the compared systems. These results demonstrate its strong performance

Table 2. Objective comparison of different systems.

Model	DNSMOS $\uparrow$	CER $\downarrow$	kD-MAE $\downarrow$	kE-MAE $\downarrow$	kP-MAE $\downarrow$	kB-RER $\downarrow$	kT-RER $\downarrow$
GroundTruth	3.616	—	0.0182	0.0133	0.0094	5.26%	8.95%
FireRedTTS3	3.599	2.89%	0.1162	0.1409	0.2326	31.29%	47.02%
Qwen3-TTS	3.642	2.99%	0.1057	0.1930	0.2291	28.36%	44.56%
CosyVoice3	3.583	3.12%	0.1126	0.1411	0.2383	24.27%	47.84%
InstCharVoice	3.686	3.58%	0.0769	0.1271	0.1857	14.04%	37.05%
- w/o Instruction	3.560	2.91%	0.0965	0.1431	0.2401	23.10%	44.08%

Table 3. Subjective evaluation results.

Model	NMOS $\uparrow$	IMOS $\uparrow$	SMOS $\uparrow$
FireRedTTS3	3.331 $\pm$ 0.266	3.036 $\pm$ 0.306	3.533 $\pm$ 0.212
Qwen3-TTS	3.277 $\pm$ 0.276	3.357 $\pm$ 0.230	3.124 $\pm$ 0.246
CosyVoice3	3.297 $\pm$ 0.264	3.402 $\pm$ 0.253	3.374 $\pm$ 0.227
InstCharVoice	3.528 $\pm$ 0.219	3.602 $\pm$ 0.212	3.173 $\pm$ 0.260

Table 4. Ablation study results.

Model	kD-MAE $\downarrow$	kE-MAE $\downarrow$	kP-MAE $\downarrow$	kB-RER $\downarrow$	kT-RER $\downarrow$
Full	0.0769	0.1271	0.1857	14.04%	37.05%
- w/o KP	0.0805	0.1383	0.1899	16.37%	37.82%
- w/o GLW	0.0859	0.1389	0.2019	20.76%	41.82%
- w/o Both	0.0906	0.1583	0.2132	21.54%	42.58%

in both speech naturalness and character-level instruction following. In contrast, the other ITTS systems obtain keyword-level acoustic results close to those of the *w/o Instruction* variant, suggesting limited ability to translate natural-language instructions into precise character-level acoustic changes. InstCharVoice obtains lower SMOS than FireRedTTS3 and CosyVoice3, suggesting a potential trade-off between speaker similarity and fine-grained acoustic control. This comparison is also influenced by the additional Seed-VC conversion applied to Qwen3-TTS and FireRedTTS3. Although its CER is slightly higher, the gap from the best result of 2.89% is only 0.69 percentage points, indicating that speech intelligibility remains competitive. Overall, InstCharVoice improves naturalness and localized instruction following while largely preserving intelligibility, with speaker similarity remaining an area for improvement.

3.3. Ablation Study

We investigate the contributions of keyword prediction (KP) and grounding-aware loss weighting (GLW). The *w/o KP* variant removes the auxiliary keyword prediction head, while *w/o GLW* disables grounding-aware attribute loss weighting. The *w/o Both* variant removes both components while retaining instruction conditioning and five-dimensional acoustic attribute prediction. For each variant, we select the checkpoint with the lowest validation loss.

As shown in Table 4, removing either component degrades keyword-level acoustic control across all five attributes. Removing GLW causes a larger degradation than removing KP, indicating that grounding-aware loss weighting has a greater impact on localized control. *w/o Both* yields the highest errors across all metrics, supporting the contribution of both components to the full model.

Table 5. Character-level control results.

Model	D-MAE $\downarrow$	E-MAE $\downarrow$	P-MAE $\downarrow$	B-ER $\downarrow$	T-RER $\downarrow$
WordVoice	0.0348	0.0484	0.1093	12.72%	24.29%
InstCharVoice	0.0393	0.0544	0.1065	12.14%	21.87%

3.4. Character-Level Control Comparison

We further compare InstCharVoice with WordVoice under explicit character-level control. Natural-language instructions are removed, and both models receive the same text, reference speech, and complete character-level acoustic attributes. The remaining settings follow the main comparison. Acoustic errors are computed over all characters using the same attribute-specific metrics, except that boundary is evaluated with strict ER rather than RER because its target categories are explicitly provided.

As shown in Table 5, InstCharVoice achieves overall character-level control performance comparable to WordVoice, with slightly higher errors for duration and energy but lower errors for pitch, boundary, and tone. These differences may reflect trade-offs among the attribute-specific training objectives rather than an overall loss of controllability. The results indicate that incorporating natural-language instruction control largely preserves the original character-level acoustic control capability.

4. CONCLUSION

We presented **InstCharVoice**, a unified TTS framework that bridges natural-language instructions and explicit character-level acoustic control. An LLM-assisted annotation pipeline constructs instructions grounded in character-level acoustic evidence, while an autoregressive model predicts instruction-relevant characters and acoustic attributes before generating the corresponding speech tokens. Keyword prediction and grounding-aware loss weighting further improve localized control. Experiments demonstrate improved instruction following and keyword-level acoustic accuracy over representative ITTS systems, together with higher naturalness scores. The framework also retains character-level control performance comparable to WordVoice under explicit attribute conditioning.

The current framework focuses on five measurable acoustic attributes and does not explicitly model abstract factors such as emotion or speaking style. Although localized control is improved, gaps remain in acoustic realization accuracy, intelligibility, and speaker similarity. Evaluation under partial attribute specifications and more complex instructions also remains to be explored. Future work will investigate broader expressive representations and preference-based optimization, including reinforcement learning, to further improve instruction following and perceptual quality.

5. REFERENCES

- [1] Zhihao Du, Yuxuan Wang, Qian Chen, Xian Shi, Xiang Lv, Tianyu Zhao, Zhifu Gao, Yexin Yang, Changfeng Gao, Hui Wang, et al., “Cosyvoice 2: Scalable streaming speech synthesis with large language models,” *arXiv preprint arXiv:2412.10117*, 2024.
- [2] Yushen Chen, Zhikang Niu, Ziyang Ma, Keqi Deng, Chunhui Wang, Jian Zhao, Kai Yu, and Xie Chen, “F5-tts: A fairytale that fakes fluent and faithful speech with flow matching,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 6255–6271.
- [3] Xueru Li, Jingyuan Xing, Xiaofen Xing, Zhipeng Li, and Xiangmin Xu, “Sa-ras: Speaker-aware style retrieval augmented generation for expressive zero-shot text-to-speech synthesis,” in *Proc. Interspeech 2025*, 2025, pp. 4388–4392.
- [4] Bajian Xiang, Cheng Wen, Han Zhao, Hao Wang, Haoxu Wang, Jiawei Jin, Jiayan Cui, Jie Chen, Mengxi Nie, Tianyu Zhao, et al., “Qwen-audio-3.0-tts: Freely controllable and highly robust speech synthesis with multi-stage training paradigm,” *arXiv preprint arXiv:2607.23938*, 2026.
- [5] Guanrou Yang, Chen Yang, Qian Chen, Ziyang Ma, Wenxi Chen, Wen Wang, Tianrui Wang, Yifan Yang, Zhikang Niu, Wenrui Liu, et al., “Emovoice: Llm-based emotional text-to-speech model with freestyle text prompting,” in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 10748–10757.
- [6] Sihang Nie, Xiaofen Xing, Jingyuan Xing, Baiji Liu, and Xiangmin Xu, “Hd-ppt: Hierarchical decoding of content and prompt-preference tokens for instruction-based tts,” in *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2026, pp. 16487–16491.
- [7] Yong Ren, Jiangyan Yi, Jianhua Tao, Haiyang Sun, Zhengqi Wen, Hao Gu, Le Xu, and Ye Bai, “Ov-instructtts: Towards open-vocabulary instruct text-to-speech,” in *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2026, pp. 16482–16486.
- [8] Jingbin Hu, Huakang Chen, Linhan Ma, Dake Guo, Qirui Zhan, Wenhao Li, Haoyu Zhang, Kangxiang Xia, Ziyu Zhang, Wenjie Tian, et al., “Voicesculptor: Your voice, designed by you,” *arXiv preprint arXiv:2601.10629*, 2026.
- [9] Feiyu Shen, Kun Xie, Yichen Wu, Ziqi Dai, Yichen Han, Junjie Li, Xuelong Geng, Fenglong Xie, Lei Xie, Xu Tang, et al., “Firedtts3: Unified speech generation and editing with semantically enriched speech representations,” *arXiv preprint arXiv:2608.17492*, 2026.
- [10] Nityanand Mathur, Hamees Sayed, Wasim Madha, Apoorv Singh, Sameer Khurana, Akshat Mandloi, and Sudarshan Kamath, “How do instructions shape speech? cross-attention attribution for style-captioned text-to-speech,” *arXiv preprint arXiv:2606.20532*, 2026.
- [11] Yixuan Zhou, Xiaoyu Qin, Zeyu Jin, Shuoyi Zhou, Shun Lei, Songtao Zhou, Zhiyong Wu, and Jia Jia, “Voxinstruct: Expressive human instruction-to-speech generation with unified multilingual codec language modelling,” in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 554–563.
- [12] Sihang Nie, Jinxin Ji, Xiaofen Xing, Deyi Tuo, Chengbin Jin, Jialong Mai, and Xiangmin Xu, “Wordvoice: Explicit and decoupled multi-dimensional word-level control for llm-based tts,” *arXiv preprint arXiv:2607.06461*, 2026.
- [13] Jialong Mai, Xiaofen Xing, and Xiangmin Xu, “Magic-tts: Fine-grained controllable speech synthesis with explicit local duration and pause control,” *arXiv preprint arXiv:2604.21164*, 2026.
- [14] Zhiyuan Zhao, Lijian Lin, Ye Zhu, Kai Xie, Yunfei Liu, and Yu Li, “Lemas: Large a 150k-hour large-scale extensible multilingual audio suite with generative speech models,” *arXiv preprint arXiv:2601.04233*, 2026.
- [15] Jin Xu, Zhifang Guo, Hangrui Hu, Yunfei Chu, Xiong Wang, Jinzheng He, Yuxuan Wang, Xian Shi, Ting He, Xinfa Zhu, et al., “Qwen3-omni technical report,” *arXiv preprint arXiv:2509.17765*, 2025.
- [16] Yafeng Chen, Siqi Zheng, Hui Wang, Luyao Cheng, Tinglong Zhu, Rongjie Huang, Chong Deng, Qian Chen, Shiliang Zhang, Wen Wang, et al., “3d-speaker-toolkit: An open-source toolkit for multimodal speaker verification and diarization,” in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [17] Alex Kendall, Yarin Gal, and Roberto Cipolla, “Multi-task learning using uncertainty to weigh losses for scene geometry and semantics,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7482–7491.
- [18] Qwen, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al., “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115*, 2025.
- [19] Zhihao Du, Changfeng Gao, Yuxuan Wang, Fan Yu, Tianyu Zhao, Hao Wang, Xiang Lv, Hui Wang, Chongjia Ni, Xian Shi, et al., “Cosyvoice 3: Towards in-the-wild speech generation via scaling-up and post-training,” *arXiv preprint arXiv:2505.17589*, 2025.
- [20] Hangrui Hu, Xinfa Zhu, Ting He, Dake Guo, Bin Zhang, Xiong Wang, Zhifang Guo, Ziyue Jiang, Hongkun Hao, Zishan Guo, et al., “Qwen3-tts technical report,” *arXiv preprint arXiv:2601.15621*, 2026.
- [21] Songting Liu, “Zero-shot voice conversion with diffusion transformers,” *arXiv preprint arXiv:2411.09943*, 2024.
- [22] Berrak Sisman, Junichi Yamagishi, Simon King, and Haizhou Li, “An overview of voice conversion and its challenges: From statistical modeling to deep learning,” *IEEE/ACM transactions on audio, speech, and language processing*, vol. 29, pp. 132–157, 2020.
- [23] Chandan KA Reddy, Vishak Gopal, and Ross Cutler, “Dnsmos p. 835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors,” in *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2022, pp. 886–890.
- [24] Xian Shi, Xiong Wang, Zhifang Guo, Yongqi Wang, Pei Zhang, Xinyu Zhang, Zishan Guo, Hongkun Hao, Yu Xi, Baosong Yang, et al., “Qwen3-asr technical report,” *arXiv preprint arXiv:2601.21337*, 2026.